

Where is technician capacity being **wasted**—and what should dispatch do about it this week?

A star-schema SQL model, a formula-driven Excel operational control, Power BI measures, and a live dashboard, built on two years of synthetic HVAC/electrical dispatch data for Ridgeline Field Services.

PREPARED FOR

**Dispatch & Operations
Leadership**

REPORTING

WINDOW

**Jan 2024 –
Dec 2025**

DATASET

**37,404 job
records**

STACK

**SQL Server · Power
BI · Excel**

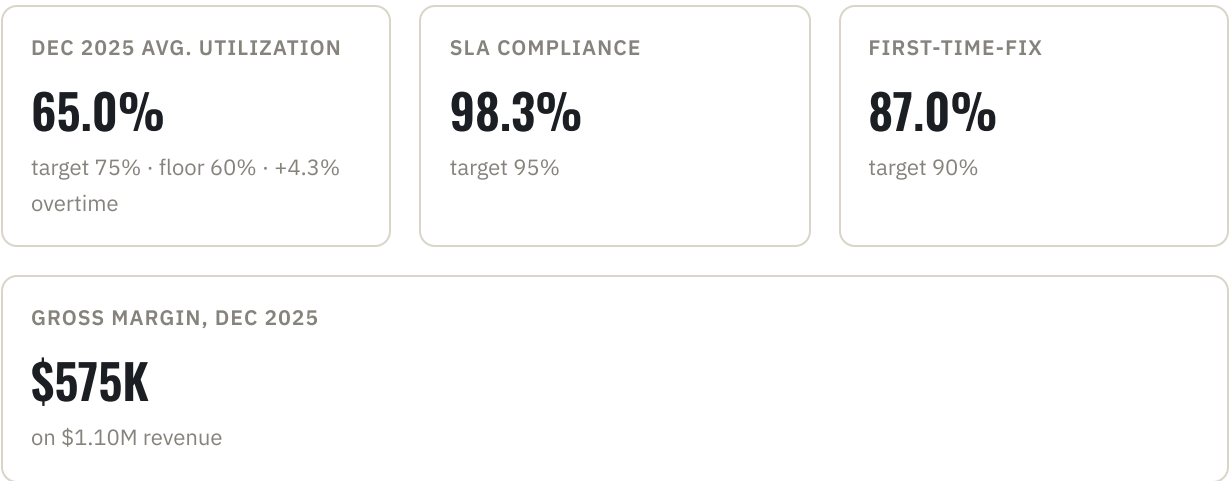
Data disclosure. Ridgeline Field Services is a fictional company built for this portfolio project. Every figure in this document is derived from synthetic data generated by `sql/02_generate_synthetic_data.sql`. No confidential, proprietary, or real-organization data is used or represented anywhere in this analysis.

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01 · Executive summary

Ridgeline Field Services dispatches HVAC and electrical technicians across five regions. Leadership's standing question was simple to ask and, until this project, expensive to answer with a spreadsheet: **where is technician capacity sitting idle, and where is it overcommitted — and what, specifically, should change?**



Three findings drive the recommendations in this report:

1. **South Metro's utilization never once overlaps with any other region's** across the full two-year window (41.1%–55.3%, vs. 59.3%+ everywhere else) — a structural staffing/dispatch imbalance, not a one-off dip.
2. **East Valley's SLA problem is invisible in the aggregate number** (97.9% overall) and only appears when urgent, 1-hour-window jobs are isolated (93.2%) — a travel-distance-driven gap that a single blended KPI would hide.
3. **Apprentice technicians run a callback rate roughly 4.7x higher than Master technicians** (21.6% vs 4.6%), concentrated enough to show up repeatedly in the priority action queue below.

All three are backed by a reusable SQL layer, cross-validated against an independently-built Excel formula layer — a cross-check that caught two real bugs during this project's build (Section 08), which is the actual argument for building the same logic twice rather than once.

02 · Business question & scope

Business question: Where is technician capacity being wasted, and which dispatch decisions should change this week?

Scope: five regions (North Metro, South Metro, East Valley, West Hills, Central District), ten HVAC/Electrical service types, 40 technicians (35 active, 5 inactive in this generation run), 300 clients, and 37,404 dispatched jobs across a 24-month window (Jan 2024–Dec 2025). Out of scope: real-time dispatch routing/GPS optimization, and multi-year demand forecasting — this is a monthly-refresh analytical model built to support a weekly dispatch-review cadence, not a live routing engine.

03 · Methodology & data validation

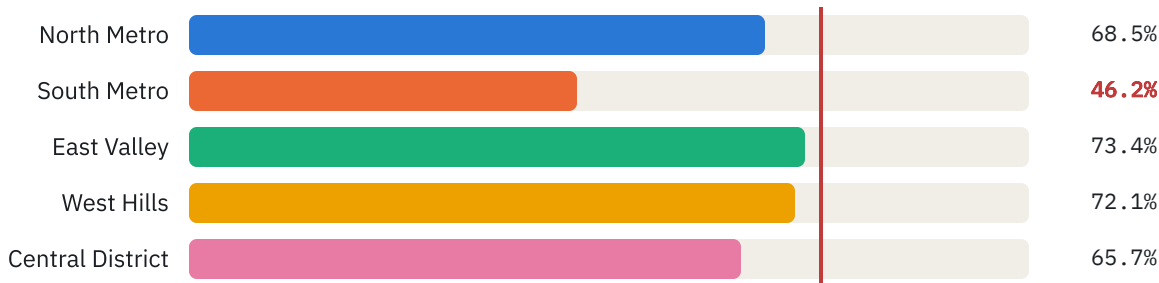
The analytical layer is a five-dimension/one-fact star schema in SQL Server (`Dim_Date` , `Dim_Region` , `Dim_Technician` , `Dim_Client` , `Dim_ServiceType` , `Fact_ServiceJobs`), plus a `Ref_KPITargets` table that centralizes every threshold used downstream — change a number there and the SQL views, the Excel dashboard, and the Power BI conditional formatting all move together.

The dataset is **synthetic by design, not by necessity** — a set-based T-SQL generator built in deliberate, documented imbalances (region workload intensity, travel-time baselines, skill-driven fix probabilities, HVAC/Electrical seasonality) so the KPI layer would have something real to find. It was then profiled — row counts, referential integrity, distribution sanity checks — before any dashboard was built on top of it. The full write-up, including a generation bug the profiling step caught (technicians dispatched before their hire date), is in `docs/data_validation_report.md`.

Every KPI is implemented as a parameterized SQL view/procedure (`sql/04_kpi_views.sql` , `sql/05_stored_procedures.sql`) and then **re-implemented independently** as live formulas in the Excel workbook — SUMIFS/AVERAGEIFS against a 37,404-row table, not a paste of query results. The two are cross-checked against each other rather than assumed to agree (Section 08).

04 · Finding 1 — Regional capacity imbalance

Utilization is measured as (job duration + travel time) actually worked, divided by available weekday capacity, per technician per month. December 2025:



Red marker = 75% target. Source: `dbo.vw_RegionKPIMonthly`, cross-checked against `KPI_Dashboard` in the Excel workbook (exact match). These are weekday capacity figures. An earlier version of this chart published the all-days ratio, which drew East Valley and West Hills above the 75% marker they do not reach.

This is not a one-month anomaly. Across all 24 months, South Metro's utilization ranges 41.1%–55.3% and **never once overlaps** with any other region's range (the next-lowest region, West Hills, never drops below 59.3%). South Metro is authorized for 8 technicians against a workload that consistently supports fewer. At the other end, East Valley averages the highest utilization of the five regions (74.2%), and several of its individual technicians post >95% utilization in a given month — an over-capacity/burnout risk (see the priority queue in Section 07).

05 · Finding 2 — Travel-driven SLA risk (a segment, not an aggregate)

SLA here is an **arrival-window commitment** (e.g. "technician arrives within 1 hour of the scheduled window"), not a multi-day completion deadline — this is how most residential/commercial field-service SLAs are actually written, and it is what makes dispatch delay and travel time SLA-relevant in the first place.

REGION	OVERALL SLA	1-HR URGENT-RESPONSE SLA	GAP
Central District	98.6%	95.3%	3.3 pts
South Metro	98.5%	95.0%	3.5 pts
North Metro	98.3%	94.6%	3.7 pts
West Hills	98.3%	94.5%	3.8 pts
East Valley	97.9%	93.2%	4.7 pts

East Valley's overall SLA rate (97.9%) looks unremarkable — it would not trip an aggregate-level alert, and would even sit within a point of the field. The gap only becomes visible when the analysis is segmented to the tight-window (1-hour) urgent-response jobs, where East Valley trails every other region and posts the widest overall-to-tight SLA drop-off in the dataset. The underlying driver is geography: East Valley's average technician travel time is roughly triple Central District's, and this project's generator deliberately encoded that difference to make exactly this kind of aggregate-vs-segment discrepancy visible — a reminder that a single blended SLA number is not sufficient evidence of dispatch health.

06 · Finding 3 — Skill-driven rework

SKILL LEVEL	FIRST-TIME-FIX	CALLBACK RATE	COMPLETED JOBS
Apprentice	78.4%	21.6%	6,477
Journeyman	88.0%	12.0%	23,845
Master	95.4%	4.6%	3,378

The gradient is clean and monotonic across all three skill tiers — Apprentices run a callback rate roughly 4.7x Master technicians' — and it is the single most consistent driver behind the "Low first-time-fix" and "High callback" flags in the priority queue below: 17 of the 25 technicians on December 2025's queue carry at least one of those two flags.

07 · The priority action queue as an operational control

Every KPI above rolls up into a single, ranked worklist — this is the artifact meant to actually change a dispatch manager's Monday, not just inform it. A technician appears here when they breach one or more thresholds in `Ref_KPITargets` ; the risk score is a simple count of breaches, and the recommended action is generated directly from which thresholds were crossed.

TECHNICIAN	REGION	UTIL.	OVERTIME	FTF	SLA	CALLBACK	RISK	ACTION
Dakota Mendez	South Metro	50.6%	7.4%	75.0%	100.0%	25.0%	3	Underutilized on weekdays but carrying 7.4% overtime; low FTF; high callback
Emerson Larsen	South Metro	56.8%	6.4%	76.7%	97.7%	23.3%	3	Underutilized on weekdays but carrying 6.4% overtime; low FTF; high callback
Hayden Weber	South Metro	48.7%	6.1%	85.7%	97.6%	14.3%	2	Underutilized on weekdays but carrying 6.1% overtime; high callback
Robin Park	West Hills	62.1%	1.3%	79.5%	95.5%	20.5%	2	Low FTF; high callback

+ 21 more, ranked by risk score — full list in the live dashboard and `excel/Ridgeline_KPI_Dashboard.xlsx`

Source: `dbo.vw_PriorityActionQueue` , reproduced independently by formula in the Excel `Priority_Action_Queue` sheet.

08 · Data quality: what was checked, and what it caught

Three anomaly types were deliberately injected into the synthetic dataset (~0.45–0.5% of completed jobs each) so the data-quality layer would have real, known-quantity issues to detect rather than a clean dataset with nothing to validate against:

CHECK	RATE	STATUS
Timestamp inversion	0.452%	PASS
Missing actuals	0.452%	PASS
Extreme duration	0.452%	PASS
Orphan dimension keys	0.000%	Clean
Inconsistent completion flags	0.000%	Clean

Excel/SQL cross-check: exact match on all three non-zero anomaly types (169/169 each) — two independently-written implementations of the same check, in two different tools, agreeing exactly.

A real bug this cross-checking caught. While validating the Priority Action Queue, the Excel formula layer and the SQL view disagreed on First-Time-Fix/SLA/Callback percentages for specific technicians — not by rounding, but by tens of percentage points. Root cause: `vw_TechnicianKPIMonthly` joined a technician's jobs without a date filter, so a technician's monthly quality metrics were silently averaged across their entire 2-year history rather than the selected month. Region-level KPIs were unaffected. The original UAT suite didn't catch it because its test fixture only ever had one month of data — nothing to leak across. The fix and a new regression test (UAT-06a/06b) are in `sql/04_kpi_views.sql` and `sql/06_uat_test_cases.sql` ; the full account is in `docs/data_validation_report.md` §5. This is the practical argument for building the same KPI logic twice, independently, rather than having Excel simply display SQL's output.

09 · Management recommendations

DISPATCH MANAGER · THIS WEEK

Rebalance South Metro against North Metro before adding headcount anywhere

South Metro's structural underutilization (averaging 47.2%, and never overlapping any other region's range across 24 months) should be addressed by demand-shifting or reallocating routing before authorizing any new hires in higher-intensity regions — the capacity gap is a distribution problem first, a headcount problem only if rebalancing doesn't close it.

REGIONAL OPS LEAD, EAST VALLEY · THIS MONTH

Review dispatch routing/travel radius specifically for 1-hour-SLA job types

Because the gap doesn't show up in East Valley's overall SLA number, this requires deliberately segmenting the review to urgent-response jobs — a blended monthly SLA report would keep missing it.

TRAINING LEAD · NEXT QUARTER

Pair Apprentice technicians with Master technicians on a rotating basis

The ~4.7x callback-rate gap between Apprentice and Master technicians is large and consistent enough to justify a structured pairing/mentorship program rather than one-off coaching — target the specific technicians flagged in the priority queue first.

ANALYTICS/BI OWNER · ONGOING

Keep the SQL and Excel KPI logic independent, and re-run the UAT suite after any view change

Section 08's bug was caught specifically because two independent implementations existed to disagree with each other. Collapsing them into one (e.g., Excel importing SQL's pre-aggregated output) would remove that safety net.

10 · Appendix — traceability, UAT, reproducibility

Requirements traceability: 10 business requirements (BR-01–BR-10), each mapped to a KPI/field, its SQL/Excel implementation, and a specific validation test — full matrix in

`docs/requirements_traceability_matrix.md`, mirrored in the Excel workbook's `Traceability_UAT` sheet.

UAT suite: 12 assertions (`sql/06_uat_test_cases.sql`) run inside a transaction against a fully isolated fixture and roll back automatically — the production dataset is never modified by running it. All 12 currently PASS, and the suite exits non-zero if any fails or if it does not run to completion. Two of them are regression tests added to close gaps that let earlier bugs through: UAT-06a/06b for the month-pooling bug in Section 08, and UAT-07a/07b for the weekday/weekend utilization defect, whose fixture UAT-01 could not reach because both its jobs fall on weekdays.

Reproducibility: the entire pipeline — schema, synthetic data generator, data-quality views, KPI views, stored procedures, UAT suite, Excel workbook builder, and this dashboard's data exports — runs from the SQL scripts and the `excel/Build_Workbook.ps1` script in this repository. See `README.md` for exact commands.

Live dashboard: an interactive version of Sections 04–07 is published and linked from the project README.

Ridgeline Field Services is a fictional company. This document and all data within it are synthetic, produced for a portfolio analytics project. Prepared with SQL Server, Power BI, Microsoft Excel, and a custom HTML dashboard.